Incident Report: Phishing Email Detection and Threat Intelligence Investigation

1. Case details

Field	Value
Case ID	`PHISH-2023-001`
Investigation date	19 October 2026
Sample date	19 September 2023, 18:35:49 UTC
Evidence filename	`sample-1.eml`
Evidence SHA-256	`4eb511271c3a8a661eca2fabfd78165b832382315c920e61b2f39e2b8c693cb2`
Case type	Public historical email sample / offline analysis
Initial verdict	**Suspicious — likely phishing**
Confidence	Medium that the message is suspicious; insufficient evidence here to call all related infrastructure currently malicious
Status	Initial analysis complete; additional reputation and DNS checks pending

2. Executive summary

The email impersonates the Banco Bradesco / Livelo rewards context and claims that 92,990 points are expiring that day. It encourages the recipient to use an HTML link to `blog1seguimentmydomaine2bra[.]me`. The sender's visible identity does not align naturally

with the Return-Path and originating host, and the receiving system recorded SPF `temperror`, DKIM none, DMARC `temperror`, and composite authentication `fail`.

These are multiple indicators that justify treating the message as suspicious. They do not, by themselves, prove that the linked site is currently active or malicious. AbuseIPDB's current page shows historical reports for the sending IP but a 0% abuse-confidence score. VirusTotal, URLScan, and current RDAP/DNS outputs must be captured and correlated before making a stronger claim.

3. Email and header analysis

Evidence screenshot: `../screenshots/01-email-headers.png` (capture and add).

Item	Observed evidence	Interpretation
From	`BANCO DO BRADESCO LIVELO <banco.bradesco@atendimento.com.br>`	Brand-themed display name; validate the sender domain and business context independently
Reply-To	Not present	No Reply-To mismatch can be asserted
Subject	`CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!`	Urgency and expiring-reward lure
Date	`Tue, 19 Sep 2023 18:35:49 +0000`	Historical sample
Return-Path	`root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06`	Different from the displayed sender identity; warrants investigation
Sending IP	`137.184.34.4`	Present in `Authentication-Results`, `Received`, and `X-Sender-IP`
SMTP mail-from	`ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06`	Host-like identity; SPF evaluation had a temporary error
SPF	`temperror`	Header reports DNS timeout; not a definitive SPF `fail`
DKIM	`none`	Receiving system reports the message was not signed with DKIM
DMARC	`temperror`	The reported result is inconclusive due to a temporary error
Composite auth	`fail reason=001`	Additional authentication concern; interpret with the full mail path
Mail-filter headers	`BCL:9`, `SCL:5`	Supporting anti-spam context; not standalone

		proof
MIME	`text/html`; Base64 transfer encoding	HTML message body; decode as text for inspection, not by rendering active links
Separate attachment	None identified	No attachment hash is available for this case
Message-ID	`<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>`	Useful search key if matching mail logs exist

4. Message-content analysis

Optional evidence screenshot: `../screenshots/10-decoded-body.png` (capture locally after decoding the HTML as text; do not visit the URL).

The HTML body advertises accumulated Livelo reward points and uses an “expiring today” message to encourage immediate action. The primary call-to-action resolves to the URL shown below. External font/style resource references also appear in the HTML; these are not automatically malicious and should not be treated as primary IOCs without further evidence.

- **Primary URL (defanged):** `hxxps://blog1seguimentmydomaine2bra[.]me/`
- **Domain (defanged):** `blog1seguimentmydomaine2bra[.]me`
- **Reason for review:** The CTA destination does not obviously align with the displayed bank/rewards brand and is paired with urgency-based social engineering.

Do not click the link while conducting this investigation. If searching a public scanner, look for an existing report first and record the scan/check date.

5. Indicator register

Optional evidence screenshot: `../screenshots/11-ioc-register.png` (capture after opening the CSV in a spreadsheet editor).

Indicator type	Value (defanged where applicable)	Source	Status
URL	hxxps://blog1seguimentmydomaine2bra[.]me/	Decoded HTML body	Extracted; reputation check pending
Domain	blog1seguimentmydomaine2bra[.]me	URL in HTML body	Extracted; VirusTotal / URLScan / RDAP / DNS pending
IPv4	137.184.34.4	Authentication-Results, Received, X-Sender-IP	Extracted; AbuseIPDB checked, see below
Sender	banco.bradesco@atendimento.com.br	From header	Extracted; legitimacy requires independent validation
Return-Path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06	Return-Path header	Extracted; suspicious context
Message-ID	<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>	Message-ID header	Extracted; possible mail-log search key
Email file SHA-256	4eb511271c3a8a661eca2fabfd78165b832382315c920e61b2f39e2b8c693cb2	Local sample-1.eml file	Computed; identifies the email evidence file, not an attachment

6. Threat-intelligence results

6.1 AbuseIPDB — checked 19 October 2026

- **IP:** 137.184.34.4
- **Abuse confidence:** 0% (Low Risk / Neutral Activity on the page at check time)
- **Reports:** 13
- **Reporters:** 9
- **Latest report:** approximately five months before the check (the page showed no reports in the last 60 days)

- **Context shown:** DigitalOcean, LLC; data center / web hosting / transit
- **Interpretation:** The history included port scanning, probing, and web-application attack categories. These historical reports are not specifically linked to this email or URL. Do not classify the IP as malicious based on this page alone.
- **Reference:** https://www.abuseipdb.com/check/137.184.34.4
- **Evidence screenshot:** `../screenshots/04-abuseipdb-ip.png` (capture and add)

6.2 VirusTotal — pending manual capture

Search the IP, domain, and full URL separately. Record the report URL, detection ratio, last-analysis time, related objects, and any community/vendor labels. If no report exists, write “No existing report found at time of check.” Do not convert a missing report or zero detections into a benign verdict.

- IP lookup: https://www.virustotal.com/gui/ip-address/137.184.34.4
- Domain lookup: https://www.virustotal.com/gui/domain/blog1seguimentmydomaine2bra.me
- **Evidence screenshots:** `../screenshots/02-virustotal-ip.png`, `../screenshots/03-virustotal-domain.png` (capture and add)

6.3 URLScan — pending manual capture

Search existing scans for the domain and URL. Record the scan date, page title, final URL, redirect chain, contacted domains/IPs, and any visible verdict. Avoid creating a public scan unless you understand its visibility and are comfortable disclosing the URL. Do not directly visit the target URL.

- Search: https://urlscan.io/
- **Evidence screenshot:** `../screenshots/05-urlscan-existing-results.png` (capture and add)

6.4 WHOIS / RDAP and DNS — pending manual capture

Check domain-registration records and DNS answers. Record the check date, source, registrar/creation date if visible, nameservers, and A/AAAA/MX/NS/TXT results. If a record is absent or the domain no longer resolves, report that precisely; an inactive domain today does not establish that it was harmless in 2023.

- ICANN Lookup: https://lookup.icann.org/
- DNS lookup: https://mxtoolbox.com/
- **Evidence screenshots:** `../screenshots/06-icann-rdap.png`, `../screenshots/07-dns-records.png` (capture and add)

7. MITRE ATT&CK mapping

- **Candidate technique:** T1566 — Phishing — Initial Access.
- **Potential sub-technique:** T1566.002 — Spearphishing Link only if further evidence supports a targeted spearphishing context. One email with a link does not prove a targeted campaign.
- **Observed behavior:** Brand/rewards impersonation, urgency-based lure, and a primary CTA link.
- **Evidence limitation:** This sample does not show whether a recipient clicked the link, entered credentials, downloaded content, or was compromised.

8. Assessment and verdict

Initial verdict: SUSPICIOUS — likely phishing

Confidence: Medium for the classification “suspicious”; low/insufficient confidence to claim currently active malicious infrastructure solely from the checks completed so far.

Supporting observations

1. The message uses a financial brand/rewards narrative and urgency around points expiring the same day.
2. The primary CTA domain does not obviously match the displayed brand.
3. The Return-Path/originating host differ from the displayed sender identity.
4. DKIM is absent and SPF/DMARC have temporary errors; composite authentication reports failure.
5. Current AbuseIPDB context shows historical reports but a 0% confidence score and no recent reports at check time, so it is not strong corroboration of this specific phishing event.

Evidence gaps

- VirusTotal IP/domain/URL results not captured in this report.
- URLScan existing-scan results not captured in this report.
- Current RDAP and DNS records not captured in this report.
- No attachment and no attachment hash.
- No evidence of recipient interaction, credential submission, or endpoint compromise.

Recommended next actions

- [ ] Complete dated VirusTotal IP/domain/URL lookups and save screenshots.
- [ ] Search URLScan existing records and save a screenshot or record “no existing scan found.”

- [ ] Complete ICANN/RDAP and DNS lookups and save outputs.
- [ ] Search mail logs by sender, subject, URL, sending IP, and Message-ID if organizational logs are available.
- [ ] If this were an active corporate incident, quarantine matching emails and assess click/credential exposure according to the organization's incident-response procedure.
- [ ] Block the URL/domain/IP only after evidence and business impact have been reviewed under an approved process.
- [ ] Update verdict and confidence when corroborating evidence is available.

9. Evidence-handling notes

- Original sample was preserved as `sample-1.eml`; calculated SHA-256 is listed in Section 1.
- The main URL is defanged in this report to reduce accidental navigation.
- No separate attachment part was identified.
- Public scan services may expose submitted data. Do not upload confidential email or internal indicators without authorization.
- Reputation results are time-sensitive and should always be documented with a date and source URL.

10. Sources

- Original public sample: https://github.com/BaturalpInce/Phishing-email-detection-using-LLM/blob/main/dataset/sample-1.eml
- AbuseIPDB: https://www.abuseipdb.com/check/137.184.34.4
- VirusTotal: https://www.virustotal.com/
- URLScan: https://urlscan.io/
- ICANN Lookup: https://lookup.icann.org/
- MITRE ATT&CK Phishing: https://attack.mitre.org/techniques/T1566/